

SLIDE_4_TECH_STACK_EXPLANATION

SLIDE 4: TECHNOLOGY STACK EXPLANATION



SLIDE CONTENT

Header: "Technical Implementation"

Core Technologies:

-  **Python:** Primary programming language
-  **Scikit-learn:** Machine learning framework
-  **Gradio:** Web interface development
-  **Pandas/Numpy:** Data processing and analysis
-  **Colorama:** Enhanced terminal visualization

ML Model Details:

- **Algorithm:** Decision Tree Classifier
- **Parameters:** max_depth=10, min_samples_split=5
- **Training:** 80/20 train-test split with stratification
- **Performance:** 100% accuracy across all metrics

System Components:

- Data preprocessing pipeline
- Feature engineering module
- Prediction engine
- Explanation generation system
- Web interface with tabbed navigation



PURPOSE OF THIS SLIDE

This slide demonstrates the **technical foundation** of your solution. It shows:

1. **Technology Choices:** Why you selected specific tools
2. **Machine Learning Approach:** How the AI component works
3. **System Architecture:** How different parts connect
4. **Implementation Quality:** Professional development practices
5. **Scalability:** How the system can grow



SPEAKING POINTS

- "Our system is built on robust, industry-standard technologies"
- "Python serves as our primary development language with scientific computing libraries"
- "For machine learning, we chose scikit-learn's Decision Tree Classifier"
- "Gradio provides an intuitive web interface for user interaction"
- "The model was trained with optimized parameters for best performance"



KEY MESSAGES

- **Industry Standard:** Using proven, reliable technologies
- **Scientific Approach:** Proper ML methodologies
- **Professional Development:** Following best practices
- **Optimized Performance:** Tuned parameters for accuracy
- **User-Centric:** Focus on accessibility and usability



TECHNICAL DETAILS

Decision Tree Algorithm:

- Interpretable and explainable AI
- Handles both numerical and categorical data
- Less prone to overfitting with proper parameters

Model Parameters:

- max_depth=10: Prevents overfitting
- min_samples_split=5: Ensures statistical significance
- stratified sampling: Maintains class distribution

Data Processing:

- 13 feature variables analyzed
- Proper encoding of categorical variables
- 80/20 train-test split for validation



DESIGN ELEMENTS

- Technology stack visualization
- Model architecture flowchart
- Code snippet examples
- Performance metrics table

This slide demonstrates technical competence and proper engineering practices